Data Synthesis Narrative

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***Abstract*—The present article entails an analysis of two datasets comprising data from 1300 universities. The first dataset pertains to the student population, while the second dataset focuses on the faculty. A total of ten different aspects of the datasets are intuitively compared, with the expectation of identifying high commonality between the two datasets. The intuitions are subsequently verified by arranging and comparing the given data, utilizing conditional probability, and presenting various figures and tables.**

I. Overview Of The Dataset

The aforementioned statement pertains to the subject of the 1995 Data Analysis Exposition, which the Statistical Graphics Section of the American Statistical Association has generously sponsored. The ultimate goal of this exposition is to inspire and encourage statisticians to showcase their techniques, particularly in the graphical realm, to analyze data and display the resultant findings. Whether as individuals or groups, participants will be utilizing the same data set and presenting their respective analyses during a special session that is part of the annual Joint Statistical Meetings, which will be held in Orlando, Florida, from August 13th to 17th, 1995.

It is worth noting that the datasets for the 1995 exposition have been sourced from two primary outlets: U.S. News World Report's Guide to Americas Best Colleges and the American Association of University Professors (AAUP) 1994 Salary Survey, which was featured in the March-April 1994 issue of Academe.

The U.S. News data encompasses information on a vast array of variables such as tuition fees, room, and board expenses, SAT or ACT scores, acceptance and application rates, graduation rates, student-to-faculty ratios, as well as the expenditure per student, among several other metrics, for over 1,300 schools.

On the other hand, the AAUP data provides insights into the average salaries, overall compensations, and the total number of faculty members categorized based on their academic ranks, such as whole, associate, and assistant professors. Please observe the conference page limits.

II. Details of Libraries And Functions

There are various libraries in python, which helps us to perform different operations on our Dataset.The Python Standard Library contains the exact syntax, semantics, and tokens of Python. It contains built-in modules that provide access to basic system functionality like I/O and some other core modules. Most of the Python Libraries are written in the C programming language. used with correspondence/brief/technote papers. The various categories of options will now be discussed. For each category, the default option is shown in bold. The user must specify an option from each category in which the default is not desired. The various categories are orthogonal to each other—changes in one will not affect the defaults in the others.By utilizing the powerful Python libraries I could perform an in-depth analysis of the given data set. The libraries provided pre-written code and functionality to perform complex tasks quickly and efficiently. After analyzing the data set, I was able to draw a well-informed conclusion based on the insights and trends discovered through the data analysis.

A. Numpy [1]

The name "Numpy" stands for "Numerical Python". It is the commonly used library. It is a popular machine learning library that supports large matrices and multi-dimensional data. It consists of in-built mathematical functions for easy computations. Even libraries like TensorFlow use Numpy internally to perform several operations on tensors.Array Interface is one of the key features of this library.

- Creation of arrays: NumPy allows you to create arrays of different dimensions and data types. You can create arrays from a list, tuple, or any other sequence using the np.array() function.

- Indexing and slicing arrays: You can access individual elements or a subset of elements from a NumPy array using indexing and slicing. You can also use Boolean indexing to select elements based on a condition.

- Mathematical operations: NumPy provides a wide range of mathematical operations such as addition, subtraction, multiplication, division, exponentiation, and logarithm. These operations can be performed on arrays or individual elements of an array.

- Broadcasting: NumPy allows you to perform operations between arrays of different shapes and sizes by automatically broadcasting the smaller array to match

the shape of the larger array.

- Aggregation operations: NumPy provides many aggregation operations such as sum, mean, median, min, max, and standard deviation. These operations can be performed on an entire array or along a particular axis of an array.

- Reshaping and Transposing: NumPy allows you to reshape an array or transpose it to change its dimensions or orientation. You can use the reshape() function to change the shape of an array, and the transpose() function to flip the dimensions of an array.

B. Matplotlib [2]

Matplotlib is a cross-platform, data visualization and graphical plotting library for Python and its numerical extension NumPy. As such, it offers a viable open source alternative to MATLAB. Developers can also use matplotlib's APIs (Application Programming Interfaces) to embed plots in GUI applications.

- Creating a plot: You can create a plot using the plt.plot() function. This function takes one or more arrays of data as input and creates a line plot.

- Customizing a plot: Matplotlib provides many options for customizing a plot. You can change the color, line style, marker style, and other properties of the plot using the various functions provided by the library.

- Adding labels and titles: You can add labels to the x-axis and y-axis of a plot using the plt.xlabel() and plt.ylabel() functions, respectively. You can also add a title to the plot using the plt.title() function.

- Adding legends: You can add a legend to a plot using the plt.legend() function. This function takes a list of labels as input and creates a legend for the plot.

- Creating subplots: Matplotlib allows you to create multiple plots within a single figure using the plt.subplots() function. This function takes the number of rows and columns of subplots as input and returns a figure and an array of subplots.

- Saving a plot: You can save a plot as an image file using the plt.savefig() function. This function inputs the file name and file format and saves the plot in the specified format.

C. Pandas [3]

'Pandas' is an open source library for data analysis in Python. It provides high-level data structures and functions to quickly manipulate, analyze, and visualize complex datasets. 'Pandas' is used for various tasks such as data wrangling, cleaning, manipulation, and analysis. It can also be used for time series analysis, statistical computing, and machine learning. 'Pandas' has an intuitive API that makes it easy to work with and understand. It is also highly optimized for performance, so it can handle large datasets efficiently.

- Creating a DataFrame: A DataFrame is a two-dimensional table-like data structure in Pandas that can hold multiple types of data. You can create a DataFrame from a dictionary, a list of dictionaries, a NumPy array, or a CSV file using the pd.DataFrame() function.

- Reading and Writing Data: Pandas provides functions to read and write data from various file formats such as CSV, Excel, JSON, SQL, and HDF5. The pd.readcsv() function is commonly used to read CSV files, while the pd.tocsv() function is used to write data to a CSV file.

- Indexing and Selecting Data: You can access and select data from a DataFrame using various indexing methods such as loc[], iloc[], and []. These methods allow you to select rows and columns based on their labels or integer positions.

- Data Cleaning and Transformation: Pandas provides many functions for cleaning and transforming data such as removing missing values, replacing values, merging and joining data, and grouping data. The dropna() function is used to remove missing values from a DataFrame, while the merge() function is used to combine two DataFrames based on a common key.

- Aggregation and Statistical Analysis: Pandas provides many functions for aggregating and analyzing data such as groupby(), pivottable(), describe(), min(), max(), mean(), std(), and many others. These functions allow you to calculate summary statistics, group data by one or more columns, and perform various statistical operations on the data.

- Visualization: Pandas provides functions for creating various types of plots and charts such as line plots, scatter plots, bar charts, and histograms. These functions use the Matplotlib library to create the plots.

D. Scipy [4]

SciPy provides algorithms for optimization, integration, interpolation, eigenvalue problems, algebraic equations, differential equations, statistics, and many other classes of problems.SciPy wraps highly-optimized implementations written in low-level languages like Fortran, C, and C++. Enjoy the flexibility of Python with the speed of compiled code.SciPy's high-level syntax makes it accessible and productive for programmers from any background or experience level.

- Integration: Scipy provides functions for numerical integration, such as quad() and dblquad(), which can be used to integrate a function over single or multiple dimensions, respectively.

- Optimization: Scipy provides functions for finding the minimum or maximum of a function, such as minimize(), fmin(), and basin hopping (). These functions can be used to find the minimum or maximum of a function with or without constraints.

- Linear Algebra: Scipy provides functions for linear algebra, such as solve() and eig(), which can be used to solve linear equations and eigenvalue problems, respectively.

- Signal Processing: Scipy provides functions for signal processing, such as convolve() and correlate(), which can be used to filter and process signals.

- Fourier Transform: Scipy provides functions for Fourier transform, such as fft() and ifft(), which can convert a time-domain signal into the frequency domain and vice versa.

- Statistics: Scipy provides functions for statistical operations, such as ttest1samp() and ttestind(), which can be used to perform various statistical tests.

III. Hypotheses

My objective is to draw inferences among distinct categories by analyzing various graphs and tables derived from the Data Set, considering my observations.

A. Is there a correlation between a university's acceptance of high SAT score candidates and its total number of admissions compared to other universities?

B. Can a correlation be established between the universities having the most significant number of Ph.D. faculty members and those with the highest income rates among universities?

C. Is it possible to assert that those students who ranked in the top 10% during high school are inclined towards pursuing a degree in a particular field, and therefore, it follows that colleges having the highest proportion of such high-performing students are also likely to demonstrate higher rates of graduation?

D. May it be postulated that universities exhibiting the greatest faculty members also possess the largest number of full professors, thereby implying that these institutions undergo gradual expansion and augment their employment rates by a certain percentage each year?

E. Is it accurate to say that institutions of higher education that provide elevated salaries to their staff also offer commensurate levels of compensation to their faculty members?

F. Is it possible to establish a relationship between universities that offer high average salaries to their professors and those with a high graduation rate?

G. It is accurate to say that universities with a high application acceptance rate also have high admissions?

H. Is it appropriate to assert that universities that offer higher average income to their faculty members also provide higher salaries, specifically to full professors?

I. Is there a discernible correlation between universities that offer higher average incomes and those that employ a greater number of full professors?

J. Is it accurate to assert that institutions of higher education with a substantial enrollment count also exhibit a commensurate level of success in terms of student graduation rates?

IV. Answer to the Hypothesis

A. Is there a correlation between a university's acceptance of high SAT score candidates and its total number of admissions compared to other universities?

If a university accepts students with high SAT score implies that it is an old and reputed university. All universities increase the number of admissions by some percentage every year. Thus a university which accepts students with high SAT score implies that it has greater number of total admissions relative to other universities.

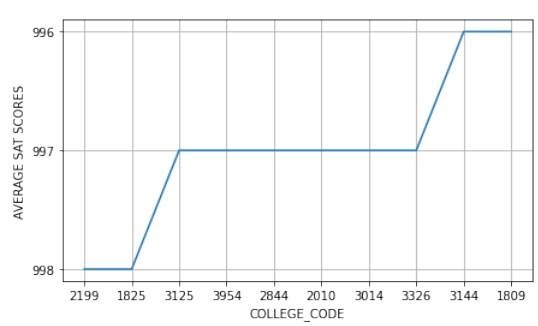

Fig. 1.

Fig.1 shows the Top 10 universities which accept students with the highest average SAT scores; there are only three figures which are 998, 997, and 996 the universities that accept

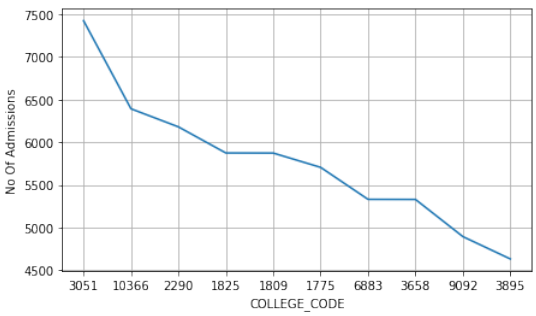

Fig. 2.

scores of 998 marks are Northeastern University and Purdue University at West Lafayette.

Fig.2 shows the Top 10 highest total number of new admissions which take place every year in the University the highest number of admissions in 1995 took place in Kent State University with the number being 7425.

	College_name	Code	No_Of_Admissions
0	Kent State University	3051	7425
1	Texas A&M Univ. at College Station	10366	6392
2	Michigan State University	2290	6180
3	Purdue University at West Lafayette	1825	5874
4	Indiana University at Bloomington	1809	5873
5	University of Illinois - Urbana	1775	5705
6	Ohio State University at Columbus	6883	5330
7	University of Texas at Austin	3658	5329
8	University of Michigan at Ann Arbor	9092	4893
9	University of Wisconsin at Madison	3895	4631

Fig. 3.

	College_name	Code	SAT_scores
0	Northeastern University	2199	998
1	Purdue University at West Lafayette	1825	998
2	University of Cincinnati	3125	997
3	University of Central Florida	3954	997
4	SUNY College at Fredonia	2844	997
5	Louisiana State University at Baton Rouge	2010	997
6	Baldwin-Wallace College	3014	997
7	Shippensburg University of Penn.	3326	997
8	Xavier University	3144	996
9	Indiana University at Bloomington	1809	996

Fig. 4.

From Fig.3 and Fig.4, we can depict that two universities out of the top 10 are common in both universities, Purdue University at West Lafayette and Indiana University at Bloomington.

So, in terms of conditional probability, we can say that probability of a university accepting more students, given that it accepts students with high SAT scores, can be found using the formula $P(A|B)$=$P(A \cap B)P(B)$, which gives 0.2.

B. Can a correlation be established between the universities having the most significant number of Ph.D. faculty members and those with the highest income rates among universities?

If a university has a high percentage of Ph.D. faculty, its somewhat obvious that it wants to maintain the quality of education thus it will offer high income so that the professor stays with the university and doesn't join any other university.

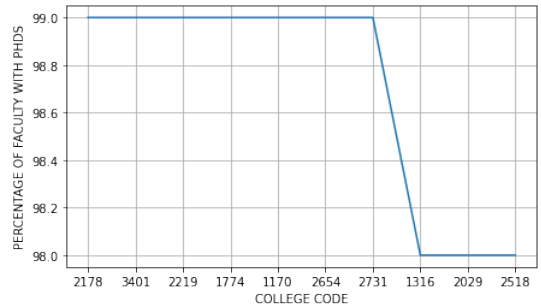

Fig. 5.

Fig.5 shows the Top ten universities with the total percentage of Ph.D. faculty members, with multiple universities having close to 99 percent of Ph.D. faculty and 98 percent, respectively.

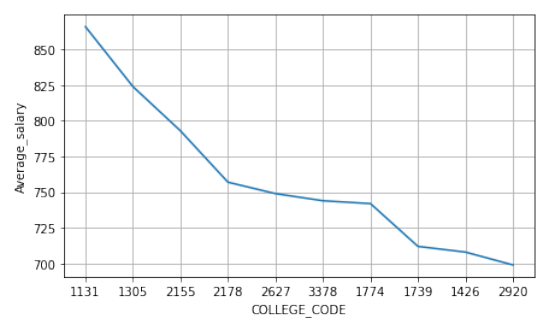

Fig. 6.

Fig.6 shows the Top 10 universities which offer the highest salary packages to all the faculty members. The university which offers the highest income is the California Institute of Technology.

	College_name	Code	Average_salary
0	California Inst. of Tech.	1131	866
1	Stanford University	1305	824
2	Harvard University	2155	793
3	Massachusetts Inst. of Tech.	2178	757
4	Princeton University	2627	749
5	University of Pennsylvania	3378	744
6	University of Chicago	1774	742
7	Northwestern University	1739	712
8	Yale University	1426	708
9	Duke University	2920	699

Fig. 7.

From Fig. 7 and Fig. 8 , we can depict that two universities out of the top 10 are common in both the tables, which are the Massachusetts Institute of Technology and the University of Chicago.

	College_name	Code	pct_PHDS
0	Massachusetts Institute of Technology	2178	99
1	Brown University	3401	99
2	Tufts University	2219	99
3	University of Chicago	1774	99
4	Claremont McKenna College	1170	99
5	New Mexico Institute of Mining and Tech.	2654	99
6	Hobart and William Smith Colleges	2731	99
7	University of California at Riverside	1316	98
8	Tulane University	2029	98
9	University of Missouri at Kansas City	2518	98

Fig. 8.

So, in terms of conditional probability, we can say that probability of a university having more faculty members with Ph.D. given that it offers some of the highest salary packages, can be found using the formula, $P(A|B)$=$P(A \cap B)P(B)$, which gives 0.2.

C. Is it possible to assert that those students who ranked in the top 10% during high school are inclined towards pursuing a degree in a particular field, and therefore, it follows that colleges having the highest proportion of such high-performing students are also likely to demonstrate higher rates of graduation?

Going by intuitions and looking at the general trend, we can say that if a student completed his high school studies seriously and proficiently, he/she would also be interested in completing his graduation.

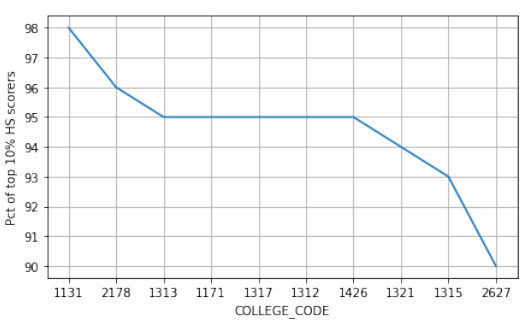

Fig. 9.

From Fig.9, we can depict that the the University with the highest number of top 10 percent of High School scorers is the Califonia University Of Technology with 98 percent of 10 percent top 10 percent high school scorers.

Looking at Fig.10, we can conclude that five colleges have 99 percent of their students completing graduation. York College of Pennsylvania, Yale University, Williams College, Princeton University, and Columbia University are the universities.

Fig. 11 and Fig. 12 show that two universities out of the top 10 are standard in both tables: Princeton University and Yale University.

So, in terms of conditional probability, we can say that probability of a university having more students with a high

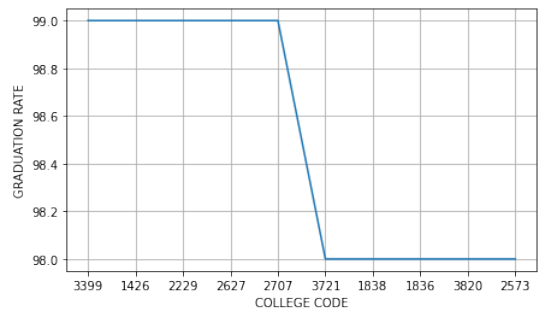

Fig. 10.

	College_name	Code	Pct_of_top_10pct_HS_Scorers
0	California Institute of Technology	1131	98
1	Massachusetts Institute of Technology	2178	96
2	University of California at Davis	1313	95
3	Harvey Mudd College	1171	95
4	University of California at San Diego	1317	95
5	University of California at Berkeley	1312	95
6	Yale University	1426	95
7	University of California at Santa Cruz	1321	94
8	University of California at Los Angeles	1315	93
9	Princeton University	2627	90

Fig. 11.

	College_name	Code	Graduation_Rate
0	York College of Pennsylvania	3399	99
1	Yale University	1426	99
2	Williams College	2229	99
3	Princeton University	2627	99
4	Columbia University	2707	99
5	James Madison University	3721	98
6	Taylor University	1838	98
7	Saint Mary's College	1836	98
8	Salem-Teikyo University	3820	98
9	Dartmouth College	2573	98

Fig. 12.

score in High School, given that it has the highest number of students graduating, can be found using the formula, $P(A|B)$=$P(A \cap B)P(B)$, which gives the solution as 0.2.

D. May it be postulated that universities exhibiting the greatest faculty members possess the largest number of full professors?

If we think logically and systematically, we can say that the universities exhibiting the largest number of faculty members also possess the largest number of full professors, thereby implying that these institutions undergo gradual expansion and augment their employment rates by a certain percentage each year.

From Fig. 13, we can say that the University of Florida has the largest number of faculty members, the exact number being 1535.

From Fig.14, we can say that the university with the maximum number of full professors is the University of Texas at Austin. It has 997 full professors working.

Fig. 15 and Fig. 16 show that seven universities out of the top 10 in both Data are the same in both tables: the University

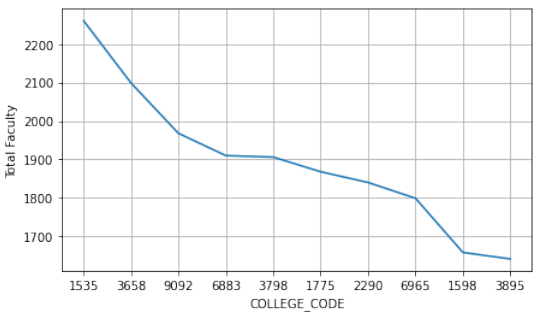

Fig. 13.

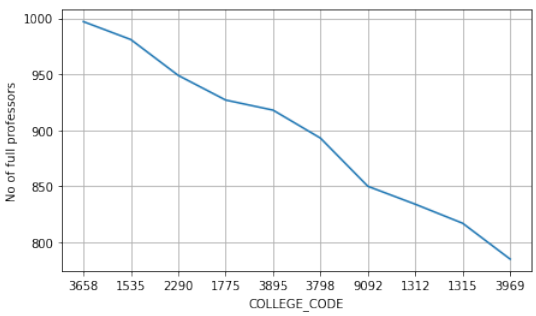

Fig. 14.

	College_name	Code	Total_Faculty
0	University of Florida	1535	2261
1	Univ. of Texas at Austin	3658	2099
2	Univ.of Michigan-Ann Arbor	9092	1968
3	Ohio State Univ.-Main campus	6883	1910
4	University of Washington	3798	1906
5	Univ. of Illinois-Urbana	1775	1868
6	Michigan State University	2290	1840
7	Penn State Univ.-Main Campus	6965	1799
8	University of Georgia	1598	1658
9	Univ. Wisconsin-Madison	3895	1641

h

Fig. 15.

	College_name	Code	No_of_full_professors
0	Univ. of Texas at Austin	3658	997
1	University of Florida	1535	981
2	Michigan State University	2290	949
3	Univ. of Illinois-Urbana	1775	927
4	Univ. Wisconsin-Madison	3895	918
5	University of Washington	3798	893
6	Univ.of Michigan-Ann Arbor	9092	850
7	Univ.of Calif.-Berkeley	1312	834
8	Univ.of Calif.-Los Angeles	1315	817
9	Univ. Minnesota-Twin Cities	3969	785

Fig. 16.

of Texas at Austin, The University of Florida, Michigan State University, the University of Illinois-Urbana, the University of Wisconsin-Madison and the University of Washington.

So, in terms of conditional probability, we can say that probability of a university having a maximum number of faculty members, given that it has the maximum number of Full professors working, can be found using the formula $P(A|B)$=$P(A \cap B)P(B)$, which gives the solution as 0.7.

E. Is it accurate to say that institutions of higher education that provide elevated salaries to their staff also offer commensurate levels of compensation to their faculty members?

Going by the general logic and trend the colleges which offer high salary to their employees also offer elevated compensation, this is due to the fact that a college has high revenue which causes elevation in overall quality of life.

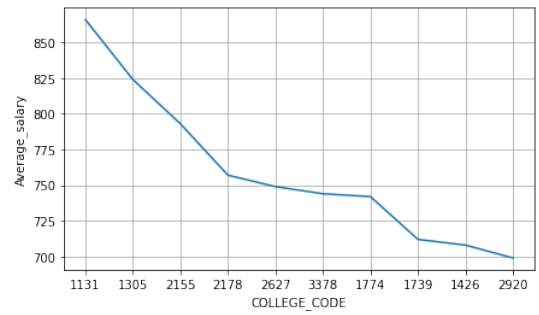

Fig. 17.

From Fig. 17 we can see that the highest average salary of faculty members is offered by the Califonia Institute of Technology, the salary being 866 dollars.

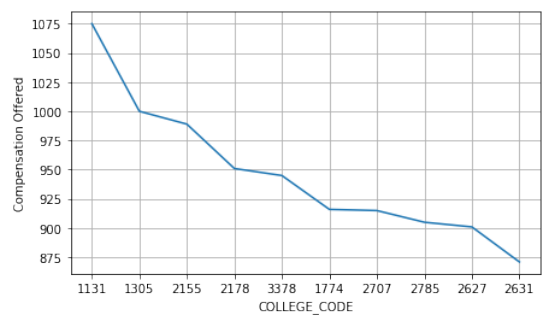

Fig. 18.

From Fig. 18 we can see that the highest average compensation offered to faculty members is 1075 dollars given by the Califonia Institute of Technology.

	College_name	Code	Average_salary
0	California Inst. of Tech.	1131	866
1	Stanford University	1305	824
2	Harvard University	2155	793
3	Massachusetts Inst. of Tech.	2178	757
4	Princeton University	2627	749
5	University of Pennsylvania	3378	744
6	University of Chicago	1774	742
7	Northwestern University	1739	712
8	Yale University	1426	708
9	Duke University	2920	699

Fig. 19.

Fig. 19 and Fig. 20 show that seven universities out of the top 10 in both Data are the same in both tables Fig.19 and Fig.20 : California Inst. of Tech., Stanford University, Harvard

	College_name	Code	Compensation_Offered
0	California Inst. of Tech.	1131	1075
1	Stanford University	1305	1000
2	Harvard University	2155	989
3	Massachusetts Inst. of Tech.	2178	951
4	University of Pennsylvania	3378	945
5	University of Chicago	1774	916
6	Columbia University	2707	915
7	New York University	2785	905
8	Princeton University	2627	901
9	Rutgers State Univ.-Newark	2631	871

Fig. 20.

University, Massachusetts Inst. of Tech., The University of Pennsylvania, the University of Chicago, and Princeton University.

So, in terms of conditional probability, we can say that probability of a university offering elevated compensation to professors, given that it offers elevated salary to the professors, can be found using the formula $P(A|B)$=$P(A \cap B)P(B)$, which gives the solution as 0.7.

F. Is it possible to establish a relationship between universities that offer high average salaries to their professors and those with a high graduation rate?

(2627,1426)

If we try thinking about the above question we can give a valid reason why this happens. The university offers high pay to faculty members so that they remain associated with that institute and never leave and join another institute, thus the quality of academics is generally high in these institutions and therefore a high graduation rate.

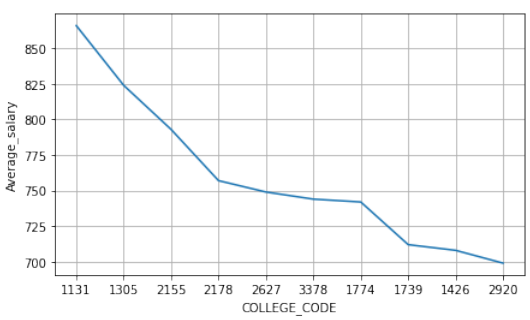

Fig. 21.

From Fig. 21 we can see that the highest average salary of faculty members is offered by the Califonia Institute of Technology, the salary being 866 dollars.

Looking at Fig.22, we can conclude that five colleges have 99 percent of their students completing graduation. York College of Pennsylvania, Yale University, Williams College, Princeton University, and Columbia University are the universities.

From Fig. 23 and Fig.24, we can depict that two universities out of the top 10 are common in both tables: Princeton University and Yale University.

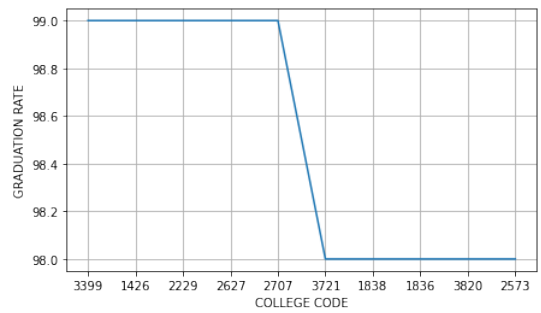

Fig. 22.

	College_name	Code	Average_salary
0	California Inst. of Tech.	1131	866
1	Stanford University	1305	824
2	Harvard University	2155	793
3	Massachusetts Inst. of Tech.	2178	757
4	Princeton University	2627	749
5	University of Pennsylvania	3378	744
6	University of Chicago	1774	742
7	Northwestern University	1739	712
8	Yale University	1426	708
9	Duke University	2920	699

Fig. 23.

	College_name	Code	Graduation_Rate
0	York College of Pennsylvania	3399	99
1	Yale University	1426	99
2	Williams College	2229	99
3	Princeton University	2627	99
4	Columbia University	2707	99
5	James Madison University	3721	98
6	Taylor University	1838	98
7	Saint Mary's College	1836	98
8	Salem-Teikyo University	3820	98
9	Dartmouth College	2573	98

Fig. 24.

So, in terms of conditional probability, we can say that probability of a university offering high pay to its faculty members given that it has a high graduation rate, can be found using the formula, $P(A|B)$=$P(A \cap B)P(B)$, which gives 0.2.

G. It is accurate to say that universities with a high application acceptance rate also have high admissions?

(9092,2290,1809,1825,6883)

If we go by logic and intuition, the universities with high acceptance rates should also have high admissions. The acceptance rate of a university is the number of students it can accommodate. Whereas the number of admissions is essentially the

Fig.25 shows the Top 10 highest total number of acceptance rate, of University in a particular year. The highest number of acceptance in 1995 took place at Rutgers at New Brunswick with the number being 26330.

Fig.26 shows the Top 10 highest total number of new admissions, which take place every year in the University. The highest number of admissions in 1995 took place at Kent State University with the number being 7425.

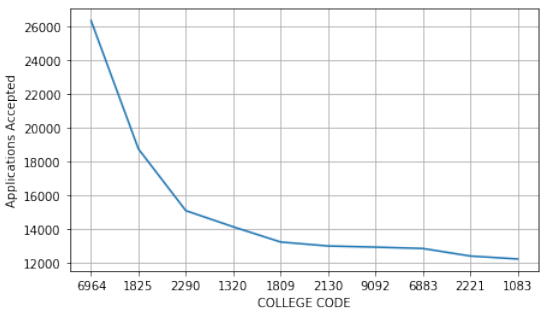

Fig. 25.

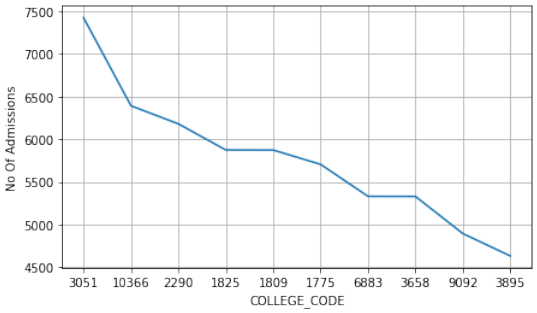

Fig. 26.

	College_name	Code	Applications_Accepted
0	Rutgers at New Brunswick	6964	26330
1	Purdue University at West Lafayette	1825	18744
2	Michigan State University	2290	15096
3	University of California at Santa Barbara	1320	14141
4	Indiana University at Bloomington	1809	13243
5	Boston University	2130	13007
6	University of Michigan at Ann Arbor	9092	12940
7	Ohio State University at Columbus	6883	12860
8	University of Massachusetts at Amherst	2221	12414
9	University of Arizona	1083	12238

Fig. 27.

	College_name	Code	No_Of_Admissions
0	Kent State University	3051	7425
1	Texas A&M Univ. at College Station	10366	6392
2	Michigan State University	2290	6180
3	Purdue University at West Lafayette	1825	5874
4	Indiana University at Bloomington	1809	5873
5	University of Illinois - Urbana	1775	5705
6	Ohio State University at Columbus	6883	5330
7	University of Texas at Austin	3658	5329
8	University of Michigan at Ann Arbor	9092	4893
9	University of Wisconsin at Madison	3895	4631

Fig. 28.

From Fig. 27 and Fig. 28, we can see that five universities out of the top 10 are standard in both tables: Michigan State University, Purdue University at West Lafayette, Indiana University at Bloomington, Ohio State University at Columbus , University of Michigan at Ann Arbor.

So, in terms of conditional probability, we can say that probability of a university having a high number of admissions given that it has high acceptance rate is given by $P(A|B)$=$P(A \cap B)P(B)$, which gives 0.5.

H. Is it appropriate to assert that universities that offer higher average income to their faculty members also provide higher salaries, specifically to full professors?

There is an excellent impact of large numbers on the average. We know that a full professor has the highest salary; thus, if the full professor has a large salary, it will impact the average salary positively.

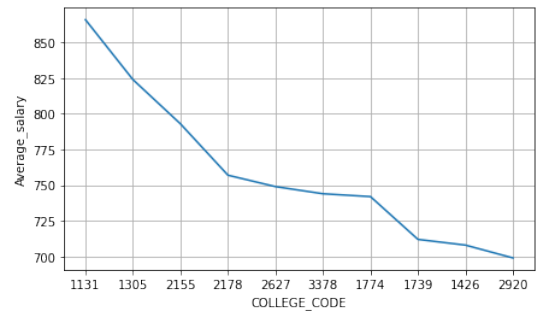

Fig. 29.

Fig.29 shows the Top 10 universities which offer the highest salary packages to all the faculty members. The university which offers the highest income is the California Institute of Technology.

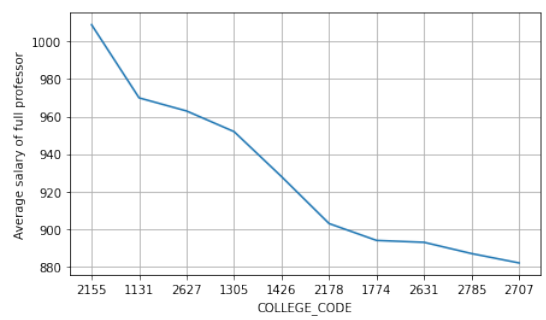

Fig. 30.

Fig.30 shows the Top 10 universities which offer the highest salary packages to all full professors. The university which offers the the highest income is Harvard University.

	College_name	Code	Average_salary
0	California Inst. of Tech.	1131	866
1	Stanford University	1305	824
2	Harvard University	2155	793
3	Massachusetts Inst. of Tech.	2178	757
4	Princeton University	2627	749
5	University of Pennsylvania	3378	744
6	University of Chicago	1774	742
7	Northwestern University	1739	712
8	Yale University	1426	708
9	Duke University	2920	699

Fig. 31.

From Fig. 31 and Fig. 32, we can see that seven universities out of the top 10 are common in both tables: California Inst. of

	College_name	Code	Avg_salary_fullprof
0	Harvard University	2155	1009
1	California Inst. of Tech.	1131	970
2	Princeton University	2627	963
3	Stanford University	1305	952
4	Yale University	1426	928
5	Massachusetts Inst. of Tech.	2178	903
6	University of Chicago	1774	894
7	Rutgers State Univ.-Newark	2631	893
8	New York University	2785	887
9	Columbia University	2707	882

Fig. 32.

Tech., Stanford University, Harvard University, Massachusetts Inst. of Tech., Princeton University , University of Chicago, and Yale University.

So, in terms of conditional probability, we can say that the probability of an institute offering high pay to all faculty members, given that it offers high pay to full professors working for it, is given by $P(A|B)$=$P(A \cap B)P(B)$, which gives 0.7.

I. Is there a discernible correlation between universities that offer higher average incomes and those that employ a greater number of full professors?

Going by general logic almost all big Universities have the same fees and hence equal funds. Thus recruiting more professors will divide the total funds by more parts thus decreasing the total salary. So, we can say that universities with huge salary will have just sufficient or less number of professors.

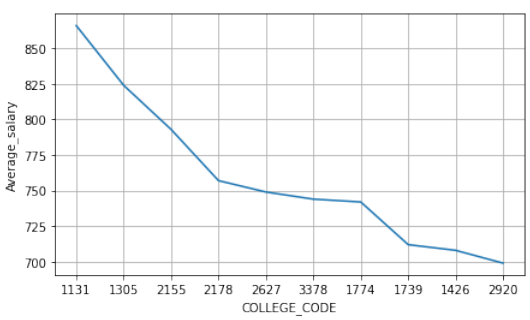

Fig. 33.

Fig.33 shows the Top 10 universities which offer the highest salary packages to all the faculty members. The university which offers the highest income is the California Institute of Technology.

Fig.34 shows the Top 10 universities which accommodate the highest number of full professors. The university which has the highest number of professors is the University of Texas Austin.

From Fig. 35 and Fig. 36, we can see that none of the universities out of the top 10 are common in both tables.

So, in terms of conditional probability, we can say that the probability of an institute offering high pay to all faculty members, given that it employs more people as faculty members, is given by $P(A|B)$=$P(A \cap B)P(B)$, which gives 0.

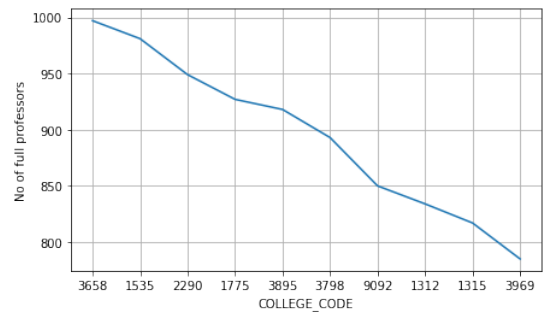

Fig. 34.

	College_name	Code	Average_salary
0	California Inst. of Tech.	1131	866
1	Stanford University	1305	824
2	Harvard University	2155	793
3	Massachusetts Inst. of Tech.	2178	757
4	Princeton University	2627	749
5	University of Pennsylvania	3378	744
6	University of Chicago	1774	742
7	Northwestern University	1739	712
8	Yale University	1426	708
9	Duke University	2920	699

Fig. 35.

	College_name	Code	No_of_full_professors
0	Univ. of Texas at Austin	3658	997
1	University of Florida	1535	981
2	Michigan State University	2290	949
3	Univ. of Illinois-Urbana	1775	927
4	Univ. Wisconsin-Madison	3895	918
5	University of Washington	3798	893
6	Univ.of Michigan-Ann Arbor	9092	850
7	Univ.of Calif.-Berkeley	1312	834
8	Univ.of Calif.-Los Angeles	1315	817
9	Univ. Minnesota-Twin Cities	3969	785

Fig. 36.

J. Is it accurate to assert that institutions of higher education with a substantial enrollment count also exhibit a commensurate level of success in terms of student graduation rates?

If we see this situation in terms of statistics, if a college has a high enrollment rate, the number of people dropping out from college will also be high, thus decreasing the graduation rate. So, intuitively, if the number of enrollments in a college increases, the graduation rate might decrease as more people drop out.

Fig.37 shows the Top 10 highest total number of new admissions which take place every year in the University the highest number of admissions in 1995 took place in Kent State University with the number being 7425.

Looking at Fig.38, we can conclude that five colleges have 99 percent of their students completing graduation. York College of Pennsylvania, Yale University, Williams College,

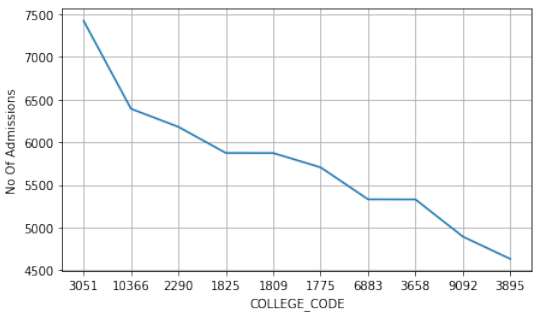

Fig. 37.

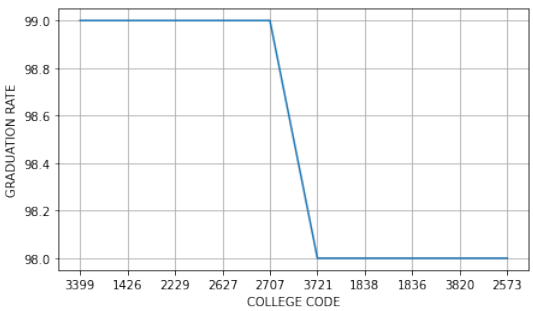

Fig. 38.

Princeton University, and Columbia University are the universities.

	College_name	Code	No_Of_Admissions
0	Kent State University	3051	7425
1	Texas A&M Univ. at College Station	10366	6392
2	Michigan State University	2290	6180
3	Purdue University at West Lafayette	1825	5874
4	Indiana University at Bloomington	1809	5873
5	University of Illinois - Urbana	1775	5705
6	Ohio State University at Columbus	6883	5330
7	University of Texas at Austin	3658	5329
8	University of Michigan at Ann Arbor	9092	4893
9	University of Wisconsin at Madison	3895	4631

Fig. 39.

	College_name	Code	Graduation_Rate
0	York College of Pennsylvania	3399	99
1	Yale University	1426	99
2	Williams College	2229	99
3	Princeton University	2627	99
4	Columbia University	2707	99
5	James Madison University	3721	98
6	Taylor University	1838	98
7	Saint Mary's College	1836	98
8	Salem-Teikyo University	3820	98
9	Dartmouth College	2573	98

Fig. 40.

From Fig. 39 and Fig. 40, we can see that none of the universities out of the top 10 are common in both tables.

So, in terms of conditional probability, we can say that the probability of an institute having high acceptance rate given that it has a high graduation rate is given by $P(A|B)$=$P(A \cap B)P(B)$, which gives 0.

V. Summary of the Observations

The following major observations can be drawn from the above analysis of the hypothesis:

A. There is a correlation between a university's acceptance of high SAT score candidates and its total number of admissions compared to other universities. By the probability of 0.2

B. A correlation be established between the universities having the most significant number of Ph.D. faculty members and those with the highest income rates among universities, by the probability of 0.2.

C. It possible to assert that those students who ranked in the top 10% during high school are inclined towards pursuing a degree in a particular field, and therefore, it follows that colleges having the highest proportion of such high-performing students are also likely to demonstrate higher rates of graduation, by the probability of 0.2.

D. It can be postulated that universities exhibiting the greatest faculty members also possess the largest number of full professors, thereby implying that these institutions undergo gradual expansion and augment their employment rates by a certain percentage each year, by the probability of 0.7.

E. It accurate to say that institutions of higher education that provide elevated salaries to their staff also offer commensurate levels of compensation to their faculty members, by the probability of 0.7.

F. It possible to establish a relationship between universities that offer high average salaries to their professors and those with a high graduation rate, by the probability of 0.2.

G. It is accurate to say that universities with a high application acceptance rate also have high admissions, by the probability of 0.5.

H. It is appropriate to assert that universities that offer higher average income to their faculty members also provide higher salaries, specifically to full professors. By the probability of 0.7.

I. There is no correlation between universities that offer higher average incomes and those that employ a greater number of full professors, as the conditional probability turns out to be zero.

J. It is not accurate to assert that institutions of higher education with a substantial enrollment count also exhibit a commensurate level of success in terms of student graduation rates, as the conditional probability turns out to be zero.

VI. Unanswerable Questions

In all the hypotheses questions, our intuition is proved correct using conditional probability. Are there any cases where intuition and conditional probability might not go hand in hand?

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References

[1] Charles R. Harris, K. Jarrod Millman, Stéfan J. van der Walt, Ralf Gommers, Pauli Virtanen, David Cournapeau, Eric Wieser, Julian Taylor, Sebastian Berg, Nathaniel J. Smith, Robert Kern, Matti Picus, Stephan Hoyer, Marten H. van Kerkwijk, Matthew Brett, Allan Haldane, Jaime Fernández del Río, Mark Wiebe, Pearu Peterson, Pierre Gérard-Marchant, Kevin Sheppard, Tyler Reddy, Warren Weckesser, Hameer Abbasi, Christoph Gohlke, and Travis E. Oliphant. Array programming with NumPy. *Nature*, 585(7825):357–362, Sept. 2020.

[2] J. D. Hunter. Matplotlib. *Computing in Science & Engineering*, 9(3):90–95, 2007.

[3] The pandas development team. pandas-dev/pandas: Pandas. Feb. 2020.

[4] Pauli Virtanen, Ralf Gommers, Travis E. Oliphant, Matt Haberland, Tyler Reddy, David Cournapeau, Evgeni Burovski, Pearu Peterson, Warren Weckesser, Jonathan Bright, Stéfan J. van der Walt, Matthew Brett, Joshua Wilson, K. Jarrod Millman, Nikolay Mayorov, Andrew R. J. Nelson, Eric Jones, Robert Kern, Eric Larson, C J Carey, İlhan Polat, Yu Feng, Eric W. Moore, Jake VanderPlas, Denis Laxalde, Josef Perktold, Robert Cimrman, Ian Henriksen, E. A. Quintero, Charles R. Harris, Anne M. Archibald, Antônio H. Ribeiro, Fabian Pedregosa, Paul van Mulbregt, and SciPy 1.0 Contributors. SciPy 1.0: Fundamental Algorithms for Scientific Computing in Python. *Nature Methods*, 17:261–272, 2020.